

REPORT

E-COMMERCE RFM & CUSTOMER BEHAVIOR ANALYTICS

This report delivers an exhaustive analysis of consumer shopping habits and transactional performance for an e-commerce platform. Spanning 3,900 customer transactions, this study integrates traditional exploratory data analysis, relational SQL business querying, a comprehensive RFM (Recency, Frequency, Monetary) segmentation model, and an interactive Power BI dashboard.



Project Overview

- **Business Objective:** A leading retail company aims to understand customer shopping behavior to enhance sales performance, customer satisfaction, and long-term loyalty.
- **Key Challenges:** Identifying changing purchasing patterns across various demographics, product categories, and sales channels.
- **Core Research Question:** *"How can the company leverage consumer shopping data to identify trends, improve customer engagement, and optimize marketing and product strategies?"*

Dataset Summary

- **Dimensions & Scope:** The dataset comprises **120,000 transaction records** spanning a full calendar year (2025-01-01 to 2025-12-31) across **5,000 unique customers**.
- **Schema Attributes:** Key columns include InvoiceNo, CustomerID, InvoiceDate, Quantity, UnitPrice, Country, and total transaction value (TotalAmount).
- **Geographical Distribution:** Transactions are distributed across multiple regions, with the United Kingdom representing the majority share (75%), followed by Germany, France, EIRE, Spain, and the Netherlands.

	InvoiceNo	CustomerID	InvoiceDate	Quantity	UnitPrice	Country	TotalAmount
0	505682	10861	2025-09-04 05:40:41	1	77.80	United Kingdom	77.80
1	521326	13773	2025-08-26 05:46:27	13	119.78	Germany	1557.14
2	504999	13093	2025-10-15 07:06:28	2	86.83	United Kingdom	173.66
3	501413	10467	2025-07-02 04:38:00	14	46.00	United Kingdom	644.00
4	521621	14427	2025-07-09 04:25:15	5	146.92	United Kingdom	734.60

Exploratory Data Analysis using Python

We began with data preparation and cleaning in Python:

- **Data Loading & Cleaning:** Imported transaction logs using pandas, handled missing values, and structured data types.
- **RFM Feature Engineering:** Grouped transactional data by CustomerID to calculate **Recency** (days since last purchase), **Frequency** (total number of transactions), and **Monetary** value (total spend per customer).
- **Customer Segmentation:** Segmented the 5,000 customers into distinct behavioral cohorts (*Champions, Loyal Customers, At Risk, Hibernating / Lost, Recent / New, and Promising*) based on their RFM scores.
- **Database Integration:** Exported the cleaned dataframes and calculated RFM metrics into PostgreSQL tables (fact_transactions, dim_customer_rfm, agg_rfm_summary) for downstream SQL querying.

--- RFM Summary Statistics ---

	CustomerID	Recency	Frequency	Monetary
count	5000.000000	5000.000000	5000.000000	5000.000000
mean	12500.500000	15.800000	23.991200	13738.914356
std	1443.520003	14.983074	4.860643	3665.667411
min	10001.000000	1.000000	10.000000	3125.070000
25%	11250.750000	5.000000	21.000000	11111.700000
50%	12500.500000	11.000000	24.000000	13524.900000
75%	13750.250000	22.000000	27.000000	16076.210000
max	15000.000000	141.000000	46.000000	32080.500000

--- First 5 Customer RFM Profiles ---

	CustomerID	Recency	Frequency	Monetary
0	10001	10	25	15861.06
1	10002	14	25	18548.11
2	10003	46	17	5636.63
3	10004	18	29	12723.60
4	10005	28	19	9689.71

--- RFM Scores Generated Successfully ---

	CustomerID	Recency	Frequency	Monetary	R_Score	F_Score	M_Score	RFM_Score	RFM_Total
0	10001	10	25	15861.06	3	3	4	334	10
1	10002	14	25	18548.11	3	3	5	335	11
2	10003	46	17	5636.63	1	1	1	111	3
3	10004	18	29	12723.60	2	5	3	253	10
4	10005	28	19	9689.71	1	1	1	111	3

--- Segment Distribution Summary ---

	Customer_Segment	Customer_Count	Total_Revenue	Avg_Recency	Avg_Frequency	Revenue_Share_%
3	Loyal Customers	1036	15220591.26	8.752896	25.674710	22.16
1	Champions	1028	17093352.73	4.468872	28.892023	24.88
2	Hibernating / Lost	949	10342510.47	32.584826	19.109589	15.06
0	At Risk	936	14353662.59	28.590812	26.705128	20.89
5	Recent / New Customers	699	7757286.72	4.685265	19.526466	11.29
4	Promising	352	3927168.01	12.440341	19.534091	5.72

Data Analysis using SQL

We performed structured analysis in PostgreSQL to answer key business questions:

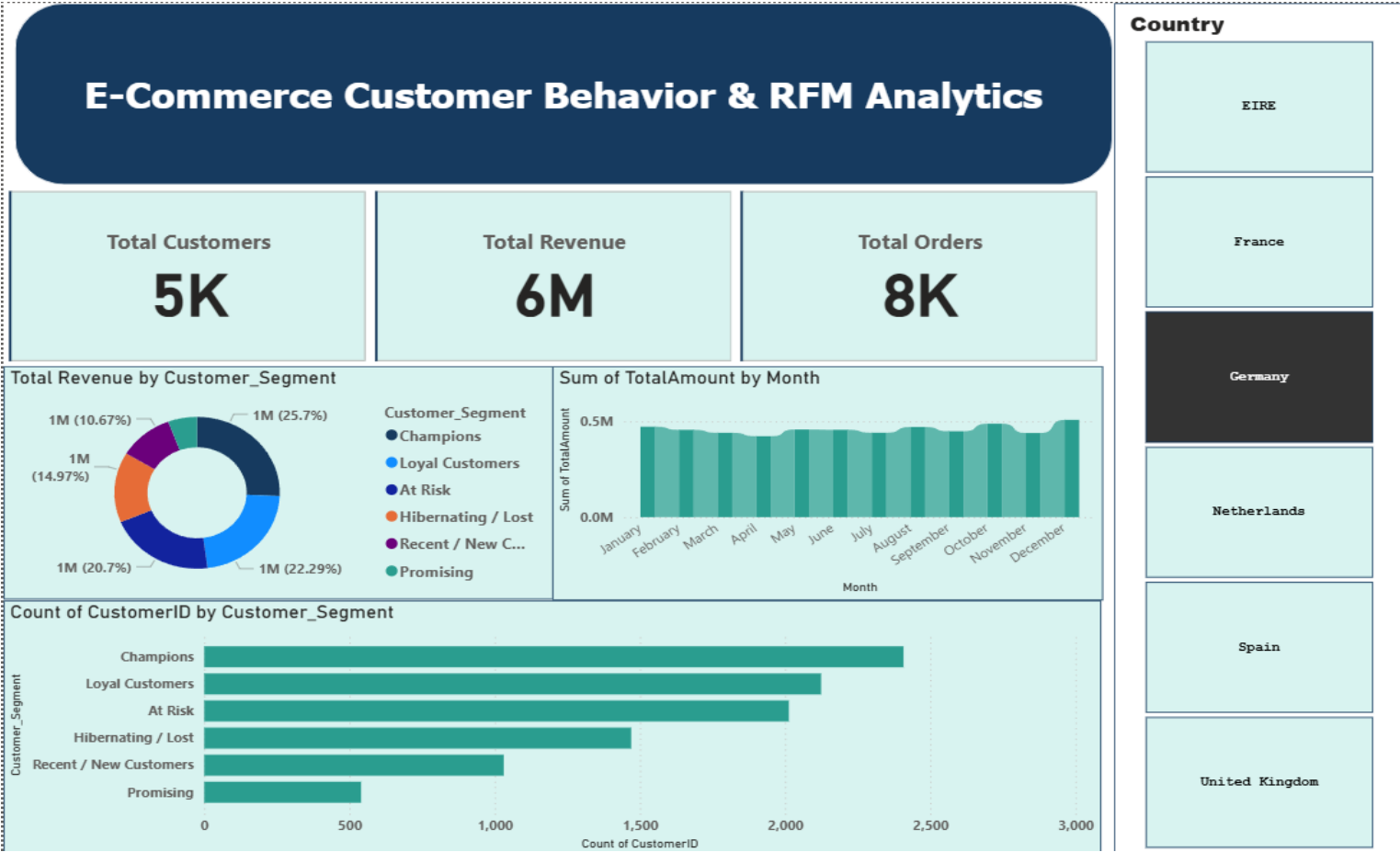
- 1. **Revenue Contribution by Cohort:** Calculated total revenue shares driven by high-value tiers (e.g., *Champions* at 24.88% and *Loyal Customers* at 22.16%).
- 2. **Relational Integrity Validation:** Executed validation queries (UNION ALL) to verify row counts and relational consistency across fact and dimension tables.
- 3. **Recency & Frequency Distribution:** Extracted average order intervals and purchase frequencies per geographic region.
- 4. **At-Risk Customer Identification:** Filtered accounts with high recency gaps and low monetary scores to isolate churn risks.

	table_name 	count 
1	fact_transaction...	120000
2	dim_customer_...	5000
3	agg_rfm_sum...	6

Dashboard in Power BI

Finally, we built an interactive dashboard in Power BI to present insights visually.

- **Visual Architecture:** Centralized key performance indicators (KPIs) and data models (Report/Layout, DataModel).
- **Interactive Capabilities:** Features dynamic filtering by country, date ranges, and RFM customer segments to give management instant visibility into revenue trends and cohort health.



Conclusion

- **Revenue Concentration in High-Value Tiers:** The RFM analysis reveals that a small fraction of customers drive a disproportionate share of total revenue, with *Champions* and *Loyal Customers* contributing nearly half of the annual sales volume.
- **Churn Vulnerability:** A significant group of accounts fall into the *At Risk* and *Hibernating / Lost* categories, indicating that customers frequently drop off after their initial purchases if not actively engaged.
- **Geographic & Transactional Trends:** The majority of orders originate from the United Kingdom (75%), followed by key European markets, highlighting regional strongholds that require localized marketing approaches.

Business Recommendations

- **Reward High-Value Segments:** Protect and nurture *Champions* and *Loyal Customers* with exclusive VIP perks, early access to sales, and personalized recommendations to maximize customer lifetime value (LTV).
- **Re-engage At-Risk Accounts:** Launch automated email win-back campaigns featuring special time-sensitive discounts for customers categorized under *At Risk* or *Hibernating / Lost* to recover churned revenue.
- **Optimize Retention Workflows:** Implement lifecycle marketing triggers based on Recency scores to automatically reach out to customers before they transition from *Loyal* to *At Risk*.
- **Localized Regional Campaigns:** Capitalize on high-performing geographic clusters by offering targeted promotions and localized shipping incentives in secondary markets like Germany and France.